DATA ANALYTICS PROJECT REPORT

ROAD ACCIDENT

DATA ANALYSIS & DASHBOARD

UK Road Safety Intelligence | 2021–2022

Tools Used	Analysis Period
Power BI | MySQL | Power Query	January 2021 – December 2022

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PROJECT OVERVIEW

Key Objectives:

- Identify accident trends over the 2021-2022 period.
- Analyze the impact of weather and road conditions on safety.
- Develop an interactive dashboard for real-time data exploration

1. Executive Summary

This report presents a comprehensive end-to-end data analytics project focused on road accident patterns in the United Kingdom spanning the years 2021 and 2022. The project was developed using three core tools: Microsoft Power BI (including Power Query for data transformation), MySQL for structured query-based analysis, and Power BI Desktop for interactive dashboard visualization.

The primary objective was to extract actionable insights from raw road accident data — covering casualty counts, accident severity, vehicle types, road conditions, geographic spread, and time-based trends — and to present them in a clear, professional, and interactive manner. The project followed a structured data analytics lifecycle: data ingestion and cleaning, exploratory data analysis via SQL, and final visualization through a Power BI dashboard.

Key findings for the Current Year (CY 2022) include a total of 195,737 casualties across 144,419 distinct accident incidents. The majority of accidents (approximately 61.9%) occurred in urban areas, and cars were overwhelmingly the most involved vehicle type, accounting for 155,804 casualties. Single carriageways remained the most accident-prone road type, and daylight conditions accounted for 73.84% of all casualties — an important counter-intuitive finding that suggests driver behaviour and traffic volume matter more than lighting.

195,737	144,419	7,135	27,045
Total CY Casualties	**Total Accidents**	**Fatal Casualties**	**Serious Casualties**
Year 2022	Distinct Incidents	All Years	Year 2022

The following sections walk through each phase of the project in detail: data preparation and cleaning, SQL-based business question analysis, and the Power BI dashboard design and insights.

2. Project Overview & Objectives

2.1 Project Background

Road accidents represent a significant public health and economic challenge across the United Kingdom. Understanding the patterns, causes, and conditions associated with road accidents is essential for policymakers, urban planners, emergency services, and the general public. This project leverages open-source UK road accident data to build a robust analytical framework.

The dataset contains accident records from 2021 and 2022, each incident characterized by attributes such as date, time, location (urban/rural), road type, speed limit, weather conditions, light conditions, vehicle type, and the number of casualties and vehicles involved.

2.2 Project Objectives

The project was designed to answer the following core business questions:

- What is the total number of casualties for the current year (CY 2022) compared to the previous year (PY 2021)?
- How do casualties break down by accident severity — Fatal, Serious, and Slight?
- Which vehicle types are most commonly involved in accidents leading to casualties?
- How do casualties vary month by month, and which periods are most dangerous?
- Which road types and urban/rural environments account for the most casualties?
- How do lighting conditions (Day vs. Night) affect the frequency of accidents?
- Which local authority areas record the highest casualty numbers?

2.3 Tools & Technologies Used

Tool / Technology	Purpose	Phase
Microsoft Power BI	Dashboard creation, data visualization	Phase 3
Power Query (M Language)	Data cleaning, transformation, shaping	Phase 1
MySQL 8.0	Data loading, SQL querying, business analysis	Phase 2
Microsoft Excel / CSV	Source data format	Phase 1
DAX (Data Analysis Expressions)	Custom measures & calculations	Phase 3

3. Phase 1 — Data Cleaning in Power Query (Power BI)

Before any analysis could be conducted, the raw dataset required thorough cleaning and transformation. This work was performed entirely within Power Query — the data transformation engine built into Microsoft Power BI. Power Query provides a visual, code-assisted interface that records every transformation step in M language, ensuring repeatability and transparency.

3.1 Data Ingestion

The raw data was provided in CSV format containing over 300,000 rows across two years (2021 and 2022). The file was imported directly into Power BI via the 'Get Data > Text/CSV' connector. Power Query automatically detected column data types and delimiters.

Upon initial load, the following initial checks were performed:

- Row count verification against the source file to ensure no records were dropped on import.
- Column header validation — ensuring all 19 expected fields were present and correctly named.
- A quick data profiling pass using the 'Column Distribution', 'Column Quality', and 'Column Profile' views in Power Query.

3.2 Handling Missing & Null Values

Data quality issues were identified in several columns. The following actions were taken:

Column	Issue Found	Action Taken
junction_control	~12% null values	Replaced nulls with 'Unknown'
carriageway_hazards	~8% null values	Replaced nulls with 'None'
weather_conditions	Minor nulls	Replaced with modal value
road_surface_conditions	Minor nulls	Replaced with 'Unknown'
vehicle_type	Inconsistent naming	Standardized text values

3.3 Duplicate Removal

Duplicate records can significantly skew casualty counts and lead to incorrect conclusions. The deduplication process followed these steps:

1. The 'accident_index' column was identified as the unique identifier for each accident incident.
2. Power Query's 'Remove Duplicates' feature was applied on the accident_index column to retain only the first occurrence of each unique accident.
3. A before-and-after row count was recorded to verify the number of duplicates removed.
4. A secondary check was performed for row-level duplicates (identical values across all columns) to catch any data entry errors.

Post-deduplication, the dataset was confirmed to contain clean, unique records ready for analysis.

3.4 Data Type Corrections

Several columns required explicit data type assignment to ensure correct sorting, filtering, and calculation behaviour in Power BI:

- accident_date: Converted from plain text (e.g., '15-06-2022') to Date type using a custom format string 'dd-MM-yyyy'.
- time: Converted from text to Time type for hour-based analysis.
- number_of_casualties and number_of_vehicles: Confirmed as Whole Number for aggregation.
- speed_limit: Converted to Whole Number to enable range-based grouping.

3.5 New Columns & Custom Groupings Created

Power Query was used to engineer several new calculated columns that were essential for dashboard-level analysis:

New Column / Group	Method	Purpose
Month Name	Date.MonthName([accident_date])	For monthly trend charts

New Column / Group	Method	Purpose
Year	Date.Year([accident_date])	To filter CY vs PY
Day of Week (clean)	Text.Proper([day_of_week])	Standardize casing
Vehicle Group	Custom conditional column	Group vehicles into Cars/Bike/Bus/Van/etc.
Light Condition Group	Custom conditional column	Group into Day/Night
Severity Group	Custom conditional column	Map severity to color codes

The Vehicle Group and Light Condition Group columns were particularly important — they consolidated over a dozen granular vehicle and lighting categories into five and two groups respectively, making the dashboard far more readable and the SQL analysis more concise.

3.6 Summary of Data Cleaning Steps

In total, the following transformations were applied in Power Query before the data was loaded into the Power BI data model:

Step No.	Transformation	Outcome
1	Source file imported via CSV connector	Raw data loaded
2	Column headers promoted from first row	Correct column names
3	Data types assigned to all 19 columns	Type-safe columns
4	Null values handled (fill, replace, or mode)	No null interference
5	Duplicate rows removed on accident_index	Clean unique dataset
6	accident_date parsed as Date type	Date hierarchy enabled
7	Vehicle Group conditional column created	Simplified categorization
8	Light Condition Group column created	Day/Night split
9	Month Name and Year extracted	Time-based filtering
10	Text casing standardized across text columns	Consistent labels

4. Phase 2 — MySQL Business Analysis

Following data cleaning, the processed CSV was loaded into a MySQL 8.0 database for structured query-based analysis. MySQL provides the ability to write precise, reproducible SQL queries that directly answer specific business questions. This phase validated the Power BI data model and allowed deeper, logic-intensive analysis that would be cumbersome to perform purely in Power Query.

4.1 Database Schema

A dedicated database named road_accident was created, containing a single primary table road_table with the following structure:

Column Name	Data Type	Description
accident_index	VARCHAR(50)	Unique identifier for each accident
accident_date	DATE	Date of accident (after conversion)
day_of_week	VARCHAR(50)	Day on which accident occurred
junction_control	VARCHAR(50)	Type of junction control present
accident_severity	VARCHAR(50)	Fatal / Serious / Slight
light_conditions	VARCHAR(50)	Lighting at time of accident
local_authority	VARCHAR(50)	Local council area
number_of_casualties	TINYINT	Count of people injured/killed
number_of_vehicles	TINYINT	Number of vehicles involved
road_type	VARCHAR(50)	Type of road (single/dual/roundabout etc.)
speed_limit	TINYINT	Posted speed limit in mph
urban_or_rural_area	VARCHAR(50)	Urban or Rural classification
weather_conditions	VARCHAR(50)	Weather at time of accident
vehicle_type	VARCHAR(50)	Type of vehicle involved

Data was loaded using MySQL's LOAD DATA INFILE command. After loading, the accident_date column was converted from TEXT to DATE format using STR_TO_DATE() with the '%d-%m-%Y' format string, followed by an ALTER TABLE to change the column's data type.

4.2 Business Question Queries & Results

The following SQL queries were written to answer each key business question. All results have been verified and documented below.

Q1: Total Casualties for Current Year (CY 2022)

This foundational KPI measures the total scale of road accident harm in the current reporting year.

Metric	SQL Function Used	Result
CY Total Casualties	SUM(number_of_casualties) WHERE YEAR = 2022	195,737

Q2: Total Distinct Accidents for CY 2022

Counting distinct accident_index values gives the number of unique accident events, as opposed to total individual casualties.

Metric	SQL Function Used	Result
CY Total Accidents	COUNT(DISTINCT accident_index) WHERE YEAR = 2022	144,419

Q3: Casualties by Severity

Breaking down casualties by severity provides a triage-level understanding of road accident impact. Fatal casualties are tracked across all years (not just 2022) to give a complete picture of the most severe outcomes.

Severity Level	Filter Applied	Total Casualties	% of CY Total
Fatal	accident_severity = 'Fatal'	7,135	3.6%
Serious	YEAR = 2022 AND severity = 'Serious'	27,045	13.8%
Slight	YEAR = 2022 AND severity = 'Slight'	165,837	84.7%

The overwhelming majority (84.7%) of CY casualties are classified as Slight. While this is positive, the 7,135 Fatal casualties remain a critical concern for road safety policy.

Q4: Casualties by Vehicle Type Group

Vehicles were grouped using a CASE WHEN clause into five meaningful categories: Agricultural, Cars, Bike, Bus, and Van. Any vehicle not fitting these categories falls into 'Other'. This analysis identifies which vehicle types are the highest risk contributors.

Vehicle Group	CY Casualties	% of Total	Vehicles Included
Cars	155,804	79.6%	Car, Taxi/Private hire car
Other	19,113	9.8%	All uncategorized vehicle types
Bike	8,809	4.5%	Motorcycles (all cc), Pedal cycle
Bus	6,042	3.1%	Coach (17+ seats), Minibus (8–16 seats)
Van	5,570	2.8%	Goods vehicles up to 7.5T
Agricultural	399	0.2%	Agricultural vehicles

Cars dominate with nearly 80% of all casualties — a direct reflection of their prevalence on UK roads. This finding underscores the importance of targeting passenger car safety through speed enforcement, driver training, and infrastructure improvements.

Q5 & Q6: Monthly Casualty Trends — CY 2022 vs PY 2021

Monthly trends reveal seasonal patterns and help identify high-risk periods requiring additional enforcement or awareness campaigns.

Month	CY 2022 Casualties	PY 2021 Casualties	Year-on-Year Change
January	13,163	18,173	-5,010 ▼
February	14,804	14,648	+156 ▲
March	16,575	17,815	-1,240 ▼
April	15,767	17,335	-1,568 ▼
May	16,775	18,852	-2,077 ▼
June	17,230	18,728	-1,498 ▼
July	17,201	19,682	-2,481 ▼
August	16,796	18,797	-2,001 ▼
September	17,500	18,456	-956 ▼
October	18,287	20,109	-1,822 ▼
November	18,439	20,975	-2,536 ▼
December	13,200	18,576	-5,376 ▼

CY 2022 consistently shows lower casualty figures than PY 2021 across nearly every month, suggesting an overall improvement in road safety over the year. The largest drop is seen in December 2022 (-5,376), possibly reflecting improved enforcement or weather-related behavioural changes. October and November remain the peak months in both years.

Q7 & Q8: Casualties by Road Type

Road type is a critical infrastructure variable. Understanding which road configurations lead to the most casualties informs highway engineering decisions.

Road Type	CY 2022 Casualties	PY 2021 Casualties
Single Carriageway	144,653	165,045
Dual Carriageway	31,912	35,456
Roundabout	12,683	14,145
One Way Street	3,499	3,890
Slip Road	2,990	3,610

Single carriageway roads account for more than 73% of all casualties in both years — a consistently dominant figure. These roads, often lacking physical separation between opposing traffic flows, are inherently more dangerous than dual carriageways. Year-on-year improvements are visible across all road types.

Q9 & Q10: Casualties by Urban vs. Rural Areas

The urban-rural split reveals fundamental differences in accident environments — urban roads have higher traffic density while rural roads often have higher speeds.

Area Type	CY 2022 %	PY 2021 %
Urban	61.9%	68.8%
Rural	38.1%	44.7%

Urban areas account for the majority of casualties in both years, driven by higher traffic volumes, more pedestrian interaction, and denser road networks. The notable decrease in urban casualties from 68.8% (2021) to 61.9% (2022) may reflect improved urban traffic management and post-COVID-19 traffic pattern normalization.

Q11 & Q12: Casualties by Light Conditions (Day vs. Night)

Light conditions at the time of an accident offer insight into driver visibility and its role in accident occurrence. A CASE WHEN grouping was used to consolidate four darkness subcategories into 'Night'.

Light Condition	CY 2022 %	PY 2021 %
Day (Daylight)	73.84%	72.22%
Night (All Darkness variants)	26.16%	27.78%

Contrary to a common assumption, the vast majority of accidents occur during daylight hours. This is explained by traffic volume — significantly more vehicles are on the road during daytime. However, night-time accidents, while fewer in number, tend to have higher severity due to reduced visibility and higher

average speeds. This finding suggests daytime road safety campaigns targeting driver distraction and speed compliance could have significant impact.

Q13: Top 10 Local Authorities by Total Casualties

This ranking identifies the geographic hotspots requiring the most urgent road safety intervention. Data was aggregated across both years for a full picture.

Rank	Local Authority	Total Casualties
1	Birmingham	8,611
2	Leeds	5,821
3	Bradford	4,431
4	Manchester	4,366
5	Liverpool	4,052
6	Cornwall	3,820
7	Sheffield	3,737
8	Kirklees	3,312
9	County Durham	3,295
10	Westminster	3,169

Birmingham ranks highest by a significant margin — nearly 50% more casualties than second-placed Leeds. This reflects both Birmingham's status as the UK's largest city outside London and its complex road network. The presence of Cornwall (a primarily rural county) in the top 10 highlights that rural roads, while producing fewer total casualties, can still generate high absolute numbers in large geographic areas.

5. Phase 3 — Power BI Dashboard & Visualizations

The final phase of the project involved designing and building an interactive Power BI dashboard that brings together all the analytical findings from Phases 1 and 2 into a single, dynamic, and visually compelling interface. The dashboard was designed with a focus on clarity, executive readability, and drill-down capability.

5.1 Data Model Design

A clean star-schema-inspired data model was constructed within Power BI:

- The cleaned road_table served as the central fact table, containing all incident records and metrics.
- A dedicated Date Table was created using DAX's CALENDAR() function to enable proper time-intelligence calculations and ensure all dates from January 2021 through December 2022 were represented, including days with no accidents.

- Relationships were established between the fact table and the Date Table via the accident_date field, enabling accurate CY vs PY comparisons.

5.2 DAX Measures Created

Custom DAX (Data Analysis Expressions) measures were written to power the KPI cards and trend comparisons. Key measures include:

Measure Name	DAX Logic / Purpose
CY Casualties	CALCULATE(SUM(number_of_casualties), YEAR = 2022)
PY Casualties	CALCULATE(SUM(number_of_casualties), YEAR = 2021)
YoY Growth %	(CY Casualties - PY Casualties) / PY Casualties
CY Fatal	CALCULATE(SUM(casualties), severity = 'Fatal', YEAR = 2022)
CY Serious	CALCULATE(SUM(casualties), severity = 'Serious', YEAR = 2022)
CY Slight	CALCULATE(SUM(casualties), severity = 'Slight', YEAR = 2022)
CY Accidents	CALCULATE(DISTINCTCOUNT(accident_index), YEAR = 2022)

5.3 Dashboard Layout & Visualizations

The dashboard was organized into a single-page layout with logically grouped sections, following a top-to-bottom reading flow:

5.3.1 Header & KPI Strip

A branded header at the top of the dashboard displays the report title and year filter. Immediately below, a row of KPI cards displays:

- Total CY Casualties with YoY % change arrow
- Total CY Accidents (distinct incidents)
- Fatal Casualties with severity badge
- Serious Casualties count
- Slight Casualties count

5.3.2 Monthly Trend Line Chart

A dual-series line chart plots CY 2022 (solid line) and PY 2021 (dashed line) casualties month by month. This visual immediately communicates the year-on-year improvement trend and highlights seasonal spikes (October–November peak). Tooltips show exact figures on hover.

5.3.3 Casualties by Road Type — Horizontal Bar Chart

A horizontal bar chart ranks road types from highest to lowest CY casualties. Single Carriageway dominates visually, making the relative risk of different road types immediately apparent. Colour coding differentiates the bars, with the highest category highlighted in the primary accent colour.

5.3.4 Urban vs. Rural Split — Donut Chart

A donut chart cleanly communicates the 61.9% Urban / 38.1% Rural split for CY 2022. The chart includes a centre label showing the total, and each segment is labelled with its percentage. A legend maps the colour coding.

5.3.5 Casualties by Vehicle Type — Tree Map / Bar Chart

Vehicle group casualties are displayed using a treemap, where the size of each rectangle corresponds to the casualty count. Cars occupy the vast majority of the treemap area, providing an instant visual impression of their dominance. Hovering reveals exact casualty numbers.

5.3.6 Day vs. Night Split — Donut Chart

A secondary donut chart mirrors the Urban/Rural visualization to show the 73.84% Day / 26.16% Night split, allowing easy side-by-side comparison of the two categorical breakdowns on the same dashboard canvas.

5.3.7 Map Visual — Geographic Accident Distribution

A filled map (choropleth) visualizes accident concentration by local authority. Darker shading indicates higher casualty counts. Birmingham, Leeds, and Manchester are clearly visible as the darkest regions. The map supports hover tooltips and can be filtered by clicking other chart elements.

5.3.8 Slicers & Filters Panel

A collapsible filter panel on the left side of the dashboard contains slicers for:

- Year (2021 / 2022 toggle)
- Accident Severity (Fatal / Serious / Slight multi-select)
- Urban or Rural Area
- Road Type

All visuals are cross-filtered — clicking any data point on any chart filters every other chart on the page, enabling rapid drill-down into specific subsets of the data. For example, clicking 'Fatal' in the severity slicer immediately updates the monthly trend, vehicle type chart, and map to show only fatal accident patterns.

5.4 Design Principles Applied

Principle	Implementation
Consistency	Uniform colour palette (deep blue primary, grey secondary) throughout
Hierarchy	KPIs at top, detailed breakdowns below — mirrors reading priority

Principle	Implementation
Minimal Clutter	Grid lines removed; axis labels kept concise; legends positioned carefully
Contextual Colour	Red used exclusively for Fatal severity; amber for Serious
Responsive Layout	All visuals aligned to a 16:9 canvas for presentation readiness
Labelling	Key data points labelled directly on charts to reduce cognitive load

6. Key Insights & Findings

The combined SQL analysis and Power BI dashboard revealed the following critical insights that have direct implications for road safety policy and public awareness:

Insight 1 — Overall Improvement in Road Safety (2021 vs. 2022)

CY 2022 recorded 195,737 total casualties versus an estimated higher figure in PY 2021 (approximately 221,000+ based on monthly totals). This represents a meaningful year-on-year reduction, suggesting that road safety measures — whether policy-driven, enforcement-based, or behavioural — are having a positive effect. December showed the largest single-month improvement (-5,376 casualties), possibly linked to targeted seasonal campaigns or weather-driven behavioural changes.

Insight 2 — Cars Are the Dominant Risk Vehicle

With 155,804 casualties (79.6% of total), cars are disproportionately involved in road accidents. While this partly reflects their sheer prevalence on UK roads, it also suggests that interventions targeting car drivers — speed cameras, distracted driving enforcement, and driver training — will yield the highest return in casualty reduction.

Insight 3 — Single Carriageway Roads Require Urgent Attention

Single carriageway roads account for over 73% of all CY casualties (144,653 of 195,737). These roads present inherent structural risks due to the lack of physical separation between opposing traffic flows and typically higher rural speeds. Infrastructure investment in central barriers, overtaking lanes, and speed management on high-risk single carriageway routes could dramatically reduce casualties.

Insight 4 — Daytime is Not Necessarily Safer

The finding that 73.84% of casualties occur in daylight conditions challenges the assumption that darkness is the primary lighting risk factor. While night-time accident severity may be higher per incident, the sheer volume of daytime accidents suggests that factors such as driver distraction, fatigue, speeding, and junction errors in busy daylight traffic are the primary casualty drivers. Road safety campaigns should give at least equal emphasis to daytime driver behaviour.

Insight 5 — Urban Areas Lead in Casualty Count; Rural Areas in Severity

Urban areas account for 61.9% of CY casualties by count, but rural casualties per incident typically involve higher speeds and therefore higher severity. The top-10 local authorities list features primarily large urban centres, confirming that urban road network complexity and high traffic volumes are key casualty drivers. Tailored interventions are needed for both environments.

Insight 6 — Birmingham is the National Hotspot

With 8,611 total casualties across both years, Birmingham records nearly 50% more casualties than second-placed Leeds (5,821). This underlines the need for targeted, city-specific road safety strategies in Birmingham, including junction redesigns, speed enforcement corridors, and pedestrian protection measures.

7. Recommendations

Based on the analytical findings, the following recommendations are proposed for road safety stakeholders including local councils, national highway agencies, and law enforcement bodies:

Priority	Recommendation	Target Area
High	Deploy average speed cameras on high-casualty single carriageway corridors	Road Infrastructure
High	Increase targeted enforcement in Birmingham, Leeds, and Manchester urban centres	Geographic Hotspots
High	Launch daytime distracted driving awareness campaigns nationally	Behavioural
Medium	Invest in central barrier installation on single carriageway A-roads	Road Infrastructure
Medium	Introduce seasonal casualty reduction targets for Oct–Nov peak months	Policy
Medium	Develop night-time rural road safety programs targeting speed and visibility	Behavioural
Low	Monitor agricultural vehicle accident trends in rural counties	Vehicle Safety
Low	Expand cycle infrastructure to reduce pedal cycle exposure on main roads	Infrastructure

8. Conclusion

This project successfully demonstrated a complete data analytics workflow — from raw CSV ingestion through structured cleaning, SQL-based business analysis, and interactive Power BI dashboard visualization — applied to a real-world road safety dataset.

The three-phase approach (Power Query cleaning → MySQL analysis → Power BI visualization) ensured that insights were built on a clean, validated data foundation, and that both technical (SQL) and non-technical (dashboard) audiences could access and interpret the findings.

The key headline finding — a meaningful year-on-year reduction in casualties from 2021 to 2022, combined with persistent risk factors around single carriageway roads, daytime accidents, and car involvement — provides a clear evidence base for prioritizing road safety interventions in the UK.

Future iterations of this project could incorporate additional data dimensions such as accident GPS coordinates for heatmap analysis, weather severity scoring for improved correlation analysis, and predictive modelling using machine learning to forecast high-risk accident zones and time periods.

9. POWERBI DASHBOARD